

1 Training

Sparse MoE(3 DP x 4 PP)

S0 S1 S2 S3



Healthy



Failed rank r

2 Detect + Repair (R1)

Failure Event
<r,t,c>

Repair Controller
(Safe-Point)

3 Guarded Policy (R2)

Score

$\Phi'(t)$

HYBRID?

YES

NO

4a Hybrid Recovery

Peers (step t)



Peer state

Replacement
 r'

Checkpoint (step c)



Checkpoint state

Two-phase

weights

resume
under barrier

optimizer

Reintegration

RECOVER

REPAIRED

BARRIER

HEALTHY

4b Checkpoint Restart



Checkpoint



Reload



Replay



Resume

5 Logs & Metrics (R3)



policy +
reason



latency



state trace



quality